

One Plot, Two Reporting Strategies

AP Statistics · Unit 5 · Topic 5.1 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **EU UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.S** — Represent bivariate quantitative data using scatterplots. [Skill 2.B]
- **EK UNC-1.S.1** — A bivariate quantitative data set consists of observations of two different quantitative variables made on individuals in a sample or population.
- **EK UNC-1.S.2** — A scatterplot shows two numeric values for each observation, one corresponding to the value on the x-axis and one corresponding to the value on the y-axis.
- **EK UNC-1.S.3** — An explanatory variable is a variable whose values are used to explain or predict corresponding values for the response variable.
- **EU DAT-1** — Regression models may allow us to predict responses to changes in an explanatory variable.
- **LO DAT-1.A** — Describe the characteristics of a scatter plot. [Skill 2.A]
- **EK DAT-1.A.1** — A description of a scatter plot includes form, direction, strength, and unusual features.
- **EK DAT-1.A.2** — The direction of the association shown in a scatterplot, if any, can be described as positive or negative.
- **EK DAT-1.A.3** — A positive association means that as values of one variable increase, the values of the other variable tend to increase. A negative association means that as values of one variable increase, values of the other variable tend to decrease.
- **EK DAT-1.A.4** — The form of the association shown in a scatterplot, if any, can be described as linear or non-linear to varying degrees.
- **EK DAT-1.A.5** — The strength of the association is how closely the individual points follow a specific pattern, e.g., linear, and can be shown in a scatterplot. Strength can be described as strong, moderate, or weak.
- **EK DAT-1.A.6** — Unusual features of a scatter plot include clusters of points or points with relatively large discrepancies between the value of the response variable and a predicted value for the response variable.

Essential question: How should several scatterplot features be weighed when descriptions conflict?

DOK-3 skill: 4.B · **FRQ pattern:** linear-label-vs-curved-pattern-description-and-consequence

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Supports Ravi with direction, form, and strength in context
- **reasoning-2** (required) — Uses 170 versus 15 mL and identifies the low 50-minute batch
- **reasoning-3** (required) — Explains how a straight line could overstate later growth

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Calling every positive association linear
- Letting one point determine the entire form
- Omitting context, strength, or the practical consequence

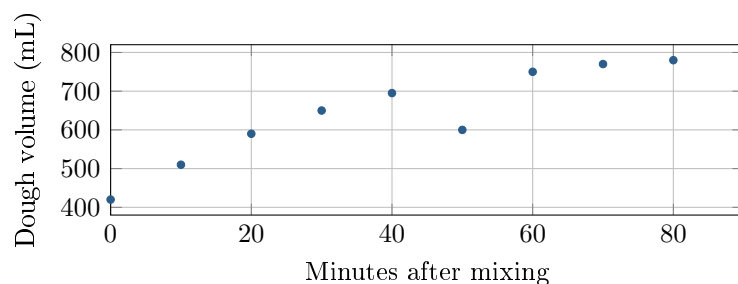
[STAR] THE PROBLEM — annotated

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

Jules: Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

Ravi: Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.

Selected measurements (minutes, mL): (0, 420), (20, 590), (70, 770), (90, 785).



FIRST TAKE — one sentence before you work the parts

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.

Key: Not graded; students revise after coordinating all features.

[BOOK] BUILD A COMPLETE SCATTERPLOT DESCRIPTION (keep on the page)

Put the explanatory variable on the x -axis and the response on the y -axis. Describe direction, form, strength, and unusual features in context. Linear form has an approximately constant rate; nonlinear form may curve or change rate. Report an unusual point without letting that one point erase the pattern followed by the rest.

DOK 1 (a) Name the explanatory and response variables, with units.

Key: Time after mixing (minutes) is explanatory; dough volume (mL) is the response.

DOK 2 (b) Find the volume increase from 0 to 20 minutes and from 70 to 90 minutes. Is the 50-minute batch above or below the nearby batches?

Key: The increases are $590 - 420 = 170$ mL and $785 - 770 = 15$ mL. The 50-minute batch at 600 mL is below the nearby 40- and 60-minute batches (695 and 750 mL).

DOK 3 (c) Whose reporting strategy is supported? Describe the direction, form, strength, and unusual point in context, using your two increases. Explain how a straight-line summary could mislead someone predicting later growth. Write 3–4 sentences.

Key: Ravi is supported: the association is fairly strong and positive, but nonlinear because most volumes rise and then level off. The 170-mL early increase versus 15 mL late supports the curve, and the 50-minute batch is unusually low compared with nearby batches. A straight-line summary could overstate later growth by suggesting that the early rate continues.